

Do Tolstoy and Hugo differ in their psychological representation of characters?

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Abstract

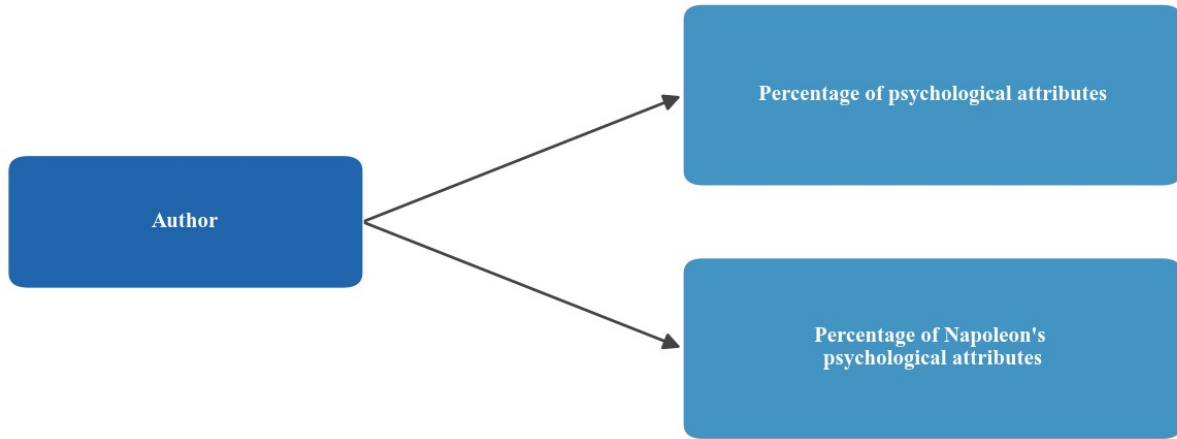
Literary critics have long distinguished Tolstoy's introspective style from Hugo's social portraiture. Using the ontological framework proposed by Bourgois et al., we try to measure this difference. We find that characters in *War and Peace* have a significantly higher proportion of psychological attributes than those of *Les Misérables* ($p = 0.003$). We see, however, that this does not hold for the specific case of Napoleon ($p = 0.528$).

1. Introduction

When analyzing characters, which are among the most important elements in understanding a literary work, a great deal of information can be gathered about both the author and the piece itself (Werner, 2021). Literary critics have long distinguished Tolstoy's introspective style from Hugo's social portraiture (Phelps, 1911). While Tolstoy renders his characters through internal monologues, Hugo tends to construct them through social functions.

Recent advances in computational literary studies offer various tools to address this question. For example, Bourgois et al. (2026) propose an ontological framework for the representation of fictional characters. Applied to the major works of Tolstoy and Hugo, respectively *War and Peace* (1869) and *Les Misérables* (1862), this framework allows us to quantify the difference in the ways the two authors construct their characters.

Do Tolstoy's characters carry a higher proportion of psychological attributes than Hugo's?



- **Prediction 1:** Characters in *War and Peace* have a higher proportion of psychological attributes than characters in *Les Misérables*.
- **Prediction 2:** Napoleon, as represented by Tolstoy in *War and Peace*, has a higher proportion of psychological attributes than Napoleon as represented by Hugo in *Les Misérables*.

2. Data

The dataset is constructed using the ontological framework proposed by Bourgois et al. (2026). First, we use the propp-fr algorithm to process each chapter of two novels, extracting character mentions, coreference chains, and attributes assigned to each character mentioned. Next, we use a trained classifier model to assign one of the 18 narratological classes to each attribute found. Finally, we group the attributes classified as affect/emotion, capability/obligation, cognition, perception, or personality traits into a psychological hyper-class. In addition, we keep only the characters that have at least 8 attributes in total in a chapter to focus on deliberate character descriptions. The final dataset contains $N = 2424$ observations (see **Figure 1**).

3. Methods

In order to test the two predictions, we performed two t-tests comparing (1) the percentage of psychological attributes depending on the author and (2) the percentage of psychological attributes of Napoleon.

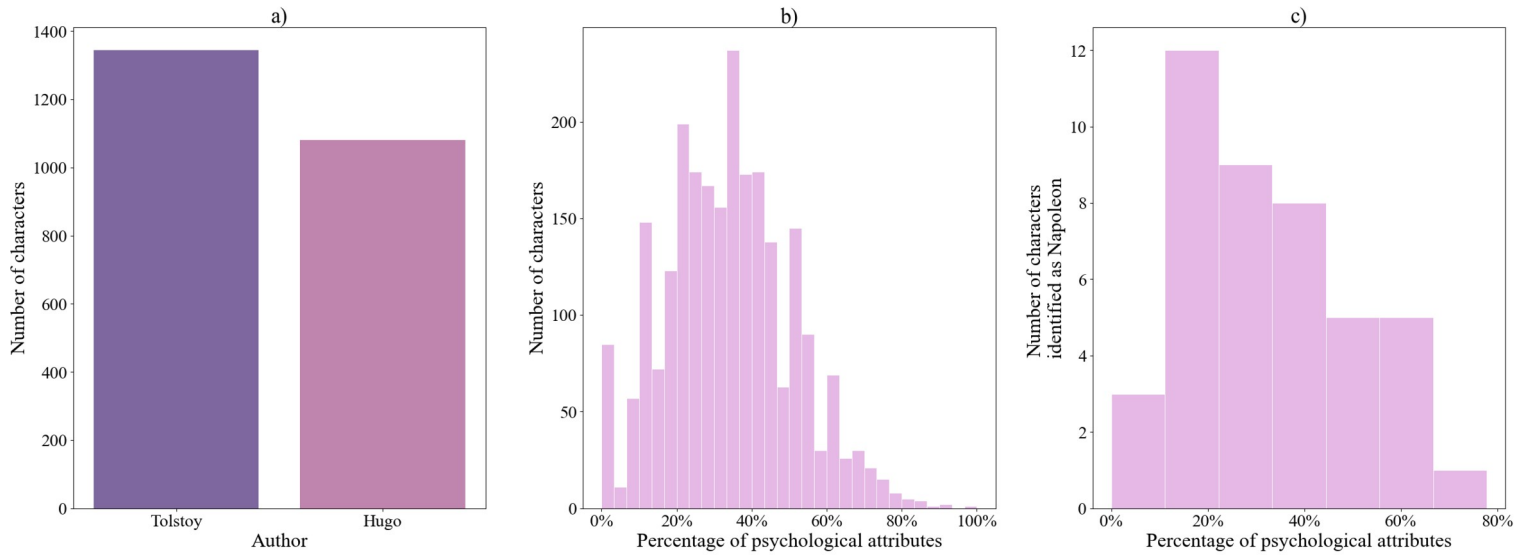


Figure 1 - Monovariate descriptions of the dataset: (a) Number of identified characters, (b) Percentage of Psychological Attributes of characters, (c) Percentage of Psychological attributes of characters identified as Napoleon

4. Results

The t-test results for our first prediction show that characters in *War and Peace* have a significantly higher proportion of psychological attributes than those of *Les Misérables* ($\mu_1 = 0.3430$, $\mu_2 = 0.3227$, $p = 0.003$, $t = 2.96$) see **Figure 2**. This suggests that Tolstoy's produces more psychologically rich characters than Hugo's. The t-test for the second prediction, however, does not give significant results ($\mu_1 = 0.3276$, $\mu_2 = 0.3690$, $p = 0.528$, $t = -0.64$) see **Figure 2**. Although the result stays insignificant, the negative t-statistic indicates that Napoleon's psychological profile is better observed in *Les Misérables*, which contradicts our prediction.

5. Conclusion

Our results indicate that characters in *War and Peace* carry a significantly higher proportion of psychological attributes than those of *Les Misérables*, confirming our hypothesis. The results differ for the specific case of Napoleon. These findings suggest that psychological interiority is indeed a measurable feature of literary works.

It should be noted, however, that our research depends heavily on the tools we have used to extract the data. The first limitation is the restriction to French-language fiction, which requires us to use the translated version of *War and Peace*. In addition, the precision of our results depends heavily on the accuracy of the propp-fr algorithm and the attributes classifier model.

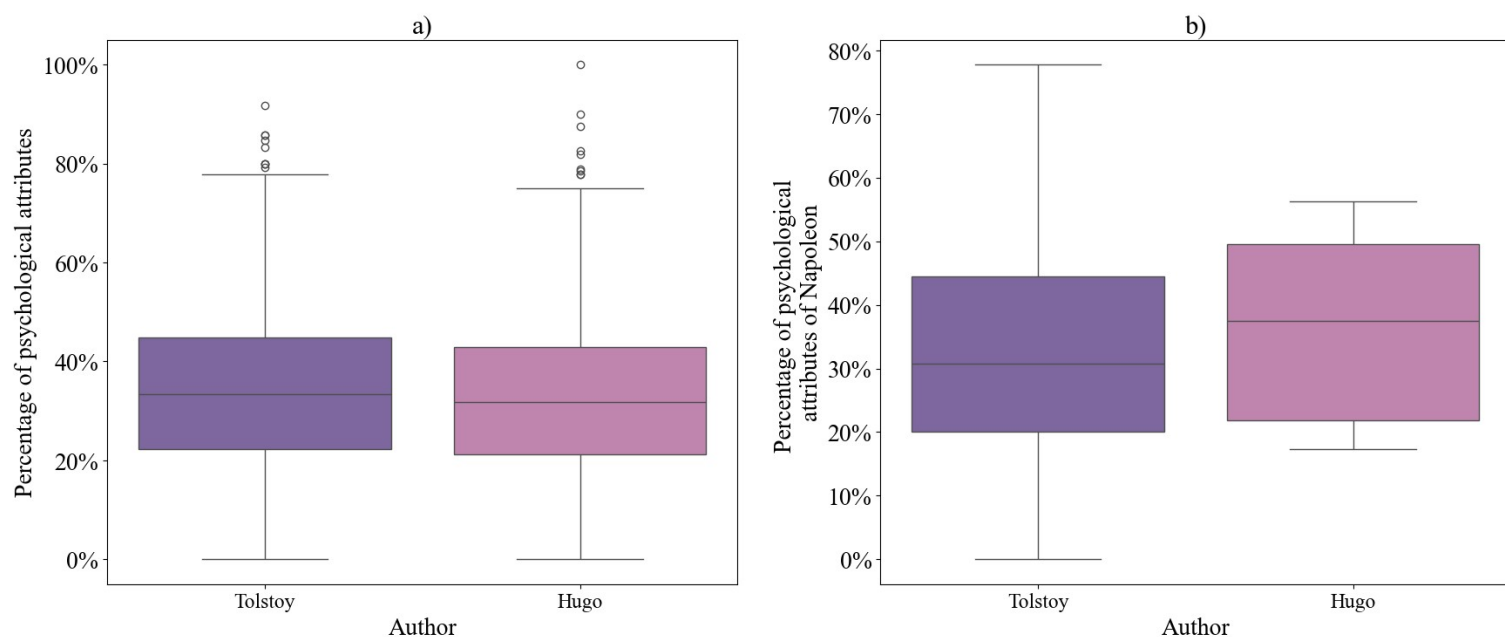


Figure 2 - Bivariate descriptions of the dataset: (a) Percentage of Psychological Attributes depending on author, (b) Percentage of Psychological Attributes of Napoleon depending on author.

6. References

Bourgeois, A., Barré, J., Seminck, O., Poibeau, T. (2026) Toward an ontological representation of fictional characters. *Computational Humanities Research*.

<https://www.cambridge.org/core/services/aop-cambridge-core/content/view/EF9F289D4EA7FC0A6ECC38BAA391DF76/S2977815826100256a.pdf/toward-an-ontological-representation-of-fictional-characters.pdf>

Phelps, W. (1911) Essays on Russian Novelists. The Macmillan Company.

<https://www.gutenberg.org/files/5996/5996-h/5996-h.htm>

Werner, C., Lüdtke, J. (2021) Empathy's Role in Different Levels of Understanding Literature: Empirical and Philosophical Perspectives, *The Journal of Aesthetics and Art Criticism*, Volume 79, Issue 2.

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